

Proof. We split the proof into two cases.

Case-I: We assume $0 < \lambda < \Lambda_\epsilon$. In view of Perron's method, we consider the solutions $\underline{u}$ and $\bar{u}$ to the equations

$$\begin{cases} \mathcal{M}_\epsilon u = \frac{\lambda}{u^{\delta(x)}} & \text{in } \Omega, \\ u > 0 & \text{in } \Omega \text{ and } u = 0 \text{ in } \mathbb{R}^2 \setminus \Omega, \end{cases} \quad (4.10)$$

and

$$\begin{cases} \mathcal{M}_\epsilon u = \frac{\bar{\lambda}}{u^{\delta(x)}} + h(x, u) & \text{in } \Omega, \\ u > 0 & \text{in } \Omega \text{ and } u = 0 \text{ in } \mathbb{R}^2 \setminus \Omega, \end{cases} \quad (4.11)$$

respectively, where $\bar{\lambda} \in (\lambda, \Lambda_\epsilon)$. For the existence of $\underline{u}$, we refer to [25]. It is immediate to show that $\underline{u}$ and $\bar{u}$ are sub and super solutions of the equation (P_λ^ϵ) , respectively. Claim: $\underline{u} \leq \bar{u}$ in Ω . To this concern, define a non-decreasing smooth function $\psi : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$\psi(t) = \begin{cases} 1 & \text{if } t \geq 1, \\ 0 & \text{if } t \leq 0. \end{cases} \quad (4.12)$$

Let $\psi_\gamma(t) = \psi(\frac{t}{\gamma})$, for $\gamma > 0$. Incorporating $\phi = \psi_\gamma(\underline{u} - \bar{u})$ in the weak formulations of (4.10) and (4.11), then subtracting one from the other, we obtain

$$\begin{aligned} B_\epsilon(\bar{u} - \underline{u}, \phi) &= \int_\Omega \left(\frac{\bar{\lambda}}{\bar{u}^{\delta(x)}} - \frac{\lambda}{\underline{u}^{\delta(x)}} \right) \phi \, dx + \int_\Omega h(x, \bar{u}) \phi \, dx \\ &\geq \lambda \int_\Omega \left(\frac{1}{\bar{u}^{\delta(x)}} - \frac{1}{\underline{u}^{\delta(x)}} \right) \phi \, dx. \end{aligned} \quad (4.13)$$

By using the monotone property of ψ , (4.13) yields

$$\begin{aligned} \lambda \int_\Omega \left(\frac{1}{\bar{u}^{\delta(x)}} - \frac{1}{\underline{u}^{\delta(x)}} \right) \phi \, dx &\leq - \int_\Omega |\nabla(\bar{u} - \underline{u})|^2 \psi'_\gamma(\underline{u} - \bar{u}) \, dx \\ &\quad + \int_{\mathbb{R}^2} \int_{\mathbb{R}^2} \frac{((\bar{u} - \underline{u})(x) - (\bar{u} - \underline{u})(y))(\phi(x) - \phi(y))}{|x - y|^{2+2s}} \, dx \, dy \\ &\leq 0, \end{aligned}$$

which reveals

$$|\{x \in \Omega : \underline{u} > \bar{u}\}| = 0.$$

Consequently, $\underline{u} \leq \bar{u}$ in Ω . Now, we define

$$\mathcal{M} := \{u \in X : \underline{u} \leq u \leq \bar{u} \text{ in } \Omega\},$$

which is a closed and convex subset of X . Define $m := \inf_{\mathcal{M}} E_{\lambda, \epsilon}(u)$. We claim that this infimum is achieved by a solution of the equation (P_λ^ϵ) . To this end, we suppose $\{u_n\}_{n \in \mathbb{N}}$ be